Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/01/2008
PI Name:	Hamins, Anthony Ph.D.		
Project Title:	THE STRUCTURE AND EXTINCTION OF LOW STRAIN RATE NON-PREMIXED FLAMES BY AN AGENT IN MICROGRAVITY		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	301-975-6662
Organization Name:	National Institute of Standards & Technology		
PI Address 1:	Building and Fire Research Laboratory		
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PI Web Page:			
City:	Gaithersburg	**State:**	MD
Zip Code:	20899-0001	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	03/31/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 3/31/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04CB07C		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A series of measurements and calculations have been undertaken in an effort improve the understanding of the suppression of low strain rate flames by an agent in microgravity. The current study extends a previous investigation by developing and comparing a two-dimensional numerical simulation of flame structure and extinction to measurements made in micro-gravity and normal gravity for methane-air non-premixed flames with Nitrogen added to the reactants. Additional measurements and calculations are planned to consider non-premixed methane and propane flames suppressed by CO2 and CF3Br. It is anticipated that a database of suppression requirements for a matrix of fuels and suppressants will be useful to NASA for determining agent mass requirements for fire protection applications. Diagnostic capabilities will be added to the current microgravity experimental rig. Task Objective: To investigate the effects of buoyancy on the structure and extinction of low strain rate flames through experimentation and computer simulation. The project is relevant from both a fire safety and a fundamental perspective. Unwanted fires typically occur due to non-premixed combustion. The agent concentration required to achieve the suppression of non-premixed flames is an important consideration in design of total flooding fire protection systems. The initial stage of an unwanted fire in a microgravity environment, such as an orbiting space platform, will likely occur at a low strain rate. A primary motivation of this proposal is to measure the agent concentrations that are required to achieve extinction of counterflowing non-premixed (or diffusion) flames at low strain rates, measurements that are not possible to perform in normal gravity due to buoyancy induced flame instabilities. For this reason, there is a gap in knowledge regarding the suppression effectiveness of agents on diffusion flames for strain rates below approximately 20 s-1.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New knowledge generated from this project could impact the search strategy for halon fire suppression replacements for fires in normal gravity, that is typical fires on Earth.
Task Progress:	The final report describes the study that has been supported through the National Aeronautics and Space Administration (NASA) microgravity grants program. This work was a continuation of a NASA microgravity grant that was active from February 1999 through September 2003. The work has used computations and measurements to investigate the structure and suppression of low strain rate non-premixed flames. This final report, on the latest grant, includes a summary of the work done on the contract and a list of publications. The work during this grant period focused on both measurements and calculations and included the following: • Continued development of the two-dimensional simulation with finite rate chemistry. [Hamins et al., 2007; Oh et al., 2007]. These publications are listed in the Bibliography section below. These calculations considered extinction of methane-air diffusion flames with N2 added to the fuel stream. The work compares calculations with our earlier measurements made in normal gravity and in microgravity using the NASA 2.2 s drop tower. • Two-dimensional calculations were completed that considered the extinction of methane-air diffusion flames with CO2 added to the fuel stream in normal and microgravity. The computations were compared to the N2 addition results described in Hamins et al., 2007; Oh et al., 2007. Complete citation information is listed in the Bibliography section below. • Flame extinction studies on the normal gravity extinction of propane were conducted. • Finally, spectroscopic measurements were conducted on key flame intermediates to support detailed modeling of radiative exchange in near-extinction flames. Wakasuki et al. [2005] developed a method to estimate the temperature dependence of absorption coefficients through an extrapolation algorithm. This publication is listed in the Bibliography section below. Spectroscopic measurements are important to understand the mechanisms of extinction associated with flames being suppressed by agents. CF3Br (Halon 1301) is one of the most important suppressants and has been used on space platforms for many years. Suppression studies often use CF3BR as a reference or baseline in considering the effectiveness of alternative agents. As part of this study, Fourier Transform Infrared Spectroscopy (FTIR) measurements were made on CF3Br at elevated temperatures. Similar spectral measurements and analysis for propane, heptane and propylene were also conducted. The manuscripts mentioned above are listed in the Bibliography section below.
Bibliography Type:	Description: (Last Updated: 02/05/2008)
Articles in Peer-reviewed Journals	Hamins A, Bundy M, Oh CB, Kim SC. "Effect of buoyancy on the radiative extinction limit of low-strain-rate nonpremixed methane–air flames." Combustion and Flame 2007 Oct;151(1-2):225-34. http://dx.doi.org/10.1016/j.combustflame.2007.05.011 , Oct-2007
Articles in Peer-reviewed Journals	Oh CB, Hamins A, Bundy M, Park J. "The two-dimensional structure of low strain rate counterflow nonpremixed-methane flames in normal and microgravity." Combustion Theory and Modelling, First Published on: 24 December 2007, forthcoming article. , Dec-2007
Articles in Peer-reviewed Journals	Wakatsuki K, Jackson GS, Kim J, Hamins A, Nyden MR, Fuss SP. "Determination of Planck mean absorption coefficients for hydrocarbon fuels." Combustion Science and Technology, in review, January 2008. , Jan-2008
Articles in Peer-reviewed Journals	Wakatsuki K, Fuss SP, Hamins A, Nyden MR. "A technique for extrapolating absorption coefficient measurements to high temperatures." Proceedings of the Combustion Institute 2005 Jan;30(1):1565-73. http://dx.doi.org/10.1016/j.proci.2004.08.078 , Jan-2005
NASA Technical Documents	Hamins A, Bundy M, Oh CB, Park J, Puri IK. "Suppression of Low Strain Rate Nonpremixed Flames by an Agent." In: Strategic Research to Enable NASA's Exploration Missions Conference and Workshop Poster Session. NASA conference publication 2004-213205/vol. 2, p. 117-126, 2004 August. NASA Glenn Research Center. , Aug-2004